MODULE 07 · DOMAIN 1 - SNOWFLAKE AI DATA CLOUD FEATURES AND ARCHITECTURE

Domain 1 drill – eight worked scenarios

No new content. Everything here comes from modules 1 to 6, in the form the exam uses.

DOMAIN 1 of 5 WEIGHT 31% MODULE 7 of 31

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How to use this drill

- **Commit to an answer before I speak.** The pause is longer here than in the teaching modules.
- **Write down why.** A right answer for the wrong reason will not survive a rephrased question.
- Domain 1 is **31%**, so roughly **31 of your 100 questions** come from this material.
- Score yourself. **Six or more of eight** suggests this domain is solid. Fewer, and the last slide says where to go back to.

> ON THE EXAM
>
> Getting one wrong here is worth more than getting it right. It tells you something a passing score would have hidden.

A team runs a small reporting job every three minutes throughout the working day. Each run executes two short queries that complete in about four seconds in total.

The warehouse is X-Small with auto-suspend set to 60 seconds. Finance has noticed the warehouse consumes far more credits than the query durations suggest it should.

What best explains the cost?

A Cloud services credits are being charged because the queries are metadata-heavy.

B X-Small warehouses have a credit floor that makes them inefficient for short queries.

C Each resume bills a 60-second minimum, and auto-suspend keeps the warehouse resuming.

D The result cache is not being used, so every run rescans the underlying tables.

◆ PAUSE AND DECIDE

What best explains the cost?

A Cloud services, metadata-heavy queries.

Cloud services is only charged above 10% of the day's warehouse credits. Warehouse usage here is high, so the allowance is generous.

B X-Small has a credit floor.

Invented. The 60-second minimum applies at every size; X-Small is simply the cheapest per second.

C The 60-second minimum, paid on every resume.

Four seconds of work bills a full minute. Twenty runs an hour pays twenty minutes for roughly 80 seconds of querying.

D Result cache not used.

Would explain slow queries, not expensive idle time. The queries complete in four seconds already.

WHERE THIS CAME FROM

Module 1. Per-second billing with a 60-second minimum on each start or resume. Fix it by lengthening auto-suspend, not by resizing.

> A regulator has asked a bank to demonstrate which users read a specific customer table over the past four months, and to show that masking was applied to analysts throughout.
>
> The bank runs Snowflake Standard Edition. An architect proposes querying QUERY_HISTORY in ACCOUNT_USAGE to produce the evidence.

What is wrong with the proposal, and what is needed?

A Nothing is wrong. QUERY_HISTORY records which columns each user read.

B QUERY_HISTORY does not give column-level read detail. ACCESS_HISTORY and masking both require Enterprise.

C ACCOUNT_USAGE retains only 30 days, so the evidence must come from INFORMATION_SCHEMA instead.

D Nothing is wrong, but Business Critical is required because the data is financial.

◆ PAUSE AND DECIDE

What is wrong with the proposal, and what is needed?

A. Nothing wrong, QUERY_HISTORY has it.

QUERY_HISTORY records statements. Column-level read detail is what ACCESS_HISTORY exists to provide.

B. Needs ACCESS_HISTORY and masking, both Enterprise.

Two separate Enterprise features. The bank cannot produce this evidence on Standard, whatever query it writes.

C. ACCOUNT_USAGE keeps only 30 days.

Wrong direction. ACCOUNT_USAGE has long retention; it is INFORMATION_SCHEMA that is the short-window view.

D. Business Critical, because financial.

'Financial data' is not a trigger. PCI cardholder data would be; ordinary customer records are not.

WHERE THIS CAME FROM

Modules 2 and 11. ACCESS_HISTORY and Column-level Security are both Enterprise. The edition is the constraint, not the query.

A 30 TB table has a wide text column that is no longer used by any query. A team drops the column, expecting the monthly storage charge for the table to fall noticeably.

Three weeks later the storage figure is essentially unchanged, although the table has had almost no DML in that period.

Why has storage not fallen?

A Time Travel is retaining the pre-drop version, and storage will fall when retention expires.

B The column was moved into Fail-safe and will be released after 7 days.

C Dropped columns are retained until the next automatic clustering pass rewrites the table.

D Dropping a column does not rewrite micro-partitions. The data stays until they are rewritten for another reason.

◆ PAUSE AND DECIDE

Why has storage not fallen?

A Time Travel is holding the old version.

Plausible for a week or two, and three weeks exceeds the 90-day maximum only on Enterprise. But even after expiry, the data would remain.

B It is in Fail-safe for 7 days.

Fail-safe applies to dropped tables and historical versions, not to a column inside a live table. Seven days have long passed.

C Released at the next clustering pass.

Invented mechanism, and the table has no clustering key mentioned. Snowflake will not recluster a table without one.

D Micro-partitions are not rewritten on a column drop.

Documented behaviour. Micro-partitions are immutable, and DROP COLUMN is a metadata change that leaves the stored data in place.

WHERE THIS CAME FROM

Module 3. Immutability cuts both ways: it is why Time Travel works, and why dropping a column reclaims nothing.

A provider has built a view that joins three internal tables and filters to a single partner's rows. They add the view to a share and grant it to the partner account.

Creating the share succeeds, but adding the view to it fails. The provider is on Enterprise Edition and the underlying tables are permanent.

What is the most likely cause?

A. The view is not secure. Only secure views can be added to a share.

B. Views cannot be shared at all. Only tables and materialized views can.

C. Views that join more than two tables cannot be shared, so it must be flattened first.

D. The provider must upgrade to Business Critical before any object can be shared.

◆ PAUSE AND DECIDE

What is the most likely cause?

A The view is not secure.

Only secure views can be shared. A non-secure view would leak its definition, and the optimiser could leak the rows it filters out.

B Views cannot be shared.

Secure views are one of the standard things you share. Sharing a filtered view is the usual pattern.

C A three-table join is too many.

Invented limit. Materialized views cannot join; ordinary and secure views can join freely.

D Business Critical is required to share.

Secure Data Sharing is Standard Edition. Business Critical adds failover, not sharing.

WHERE THIS CAME FROM

Module 4. Secure is a property, not a view type, and it is the property that makes a view shareable.

> An application needs to record customer sessions, with single-row lookups and updates by session id arriving many times per second. It requires that session ids are unique and that a foreign key to the customers table is enforced by the database.
>
> The team also wants to run analytical queries over the same session data alongside their existing warehouse tables.

Which table type fits?

A. A permanent table with a clustering key on session_id.

B. An Iceberg table with Snowflake as the catalog.

C. A hybrid table.

D. A transient table with the Search Optimization Service enabled.

◆ PAUSE AND DECIDE

Which table type fits?

A Permanent table, clustered on session_id.

Ordinary Snowflake tables do not enforce unique or foreign key constraints. A unique id is also the classic bad clustering key.

B Iceberg with Snowflake as catalog.

An analytical open table format. No enforced constraints and no row-level locking for high-frequency single-row writes.

C A hybrid table.

Enforced primary key, unique and referential constraints, indexes and row-level locking. This is exactly what Unistore is for.

D Transient plus search optimization.

Search optimization speeds point lookups but enforces nothing, and transient adds a data-loss risk nobody asked for.

WHERE THIS CAME FROM

Module 4. 'Enforced by the database' is the tell. Ordinary Snowflake tables do not enforce constraints other than NOT NULL.

A new analyst cannot connect to Snowflake from a client on the corporate network. The connection times out. Colleagues on the same network connect without difficulty, and the analyst can sign in to Snowsight in a browser.

The team suspects a proxy or firewall issue but is not certain which host or port is being blocked.

What is the most appropriate first step?

A Switch the analyst to the SQL API, which uses HTTPS and avoids proxy problems.

B Run SnowCD, the connectivity diagnostic tool, to identify what is failing.

C Add the analyst's IP address to a network policy allowed list.

D Reinstall SnowSQL, since automatic upgrades were disabled from version 1.3.0.

◆ PAUSE AND DECIDE

What is the most appropriate first step?

A. Switch to the SQL API.

Works around the symptom without diagnosing it, and changes the analyst's tooling to avoid a problem nobody has identified.

B. Run SnowCD.

SnowCD exists for exactly this. It is a diagnostic tool rather than a way to connect, and it reports which hosts and ports are unreachable.

C. Add the IP to a network policy.

A network policy blocks with an explicit denial, not a timeout, and colleagues on the same network connect fine.

D. Reinstall SnowSQL.

True that automatic upgrades were disabled from 1.3.0, but nothing suggests a version problem. A true fact that answers nothing.

WHERE THIS CAME FROM

Module 5. SnowCD is the odd one out in the connectivity table: not a way to connect, but a way to find out why you cannot.

> A security review finds that every user in an account has been calling Cortex AI functions against tables containing customer correspondence. Nobody recalls enabling Cortex, and the security team assumed it was off by default.
>
> They want only a single data science role to retain access.

What happened, and what should they do?

A Someone granted CORTEX_USER to PUBLIC. Revoke that grant from the individual users.

B Cortex inherits the ACCOUNTADMIN role by default. Restrict ACCOUNTADMIN membership.

C Cortex is enabled per warehouse. Disable it on all warehouses except the data science one.

D CORTEX_USER and USE AI FUNCTIONS are granted to PUBLIC by default. Revoke from PUBLIC, then grant to the data science role.

◆ PAUSE AND DECIDE

What happened, and what should they do?

A Someone granted it. Revoke from the users.

Nobody granted anything; it is the default. And you cannot revoke a database role from a user, because it was never granted to one.

B Cortex inherits ACCOUNTADMIN.

Invented. Access comes from a database role plus an account privilege, both held via PUBLIC.

C Enabled per warehouse.

Warehouses supply compute. They are not an access control boundary for any feature.

D Both are granted to PUBLIC by default. Revoke, then grant.

The documented default and the documented remedy. PUBLIC is automatically granted to every user and role.

WHERE THIS CAME FROM

Module 6. Nobody switched Cortex on. It was on from account creation, because PUBLIC holds both grants.

> A compliance policy requires that a particular table can be restored to any point within the past 30 days. The table is currently transient, and the account is on Standard Edition.
>
> An engineer runs ALTER TABLE … SET DATA_RETENTION_TIME_IN_DAYS = 30 and reports that the requirement is now met.

Is the requirement met, and what is actually needed?

A. No. Two changes are needed: a permanent table, and Enterprise Edition or higher.

B. Yes. Retention can be set to 30 days on any table type in any edition.

C. No. The table must become permanent, but Standard Edition already supports 30 days.

D. No. Business Critical is required, because 30-day retention is a compliance feature.

◆ PAUSE AND DECIDE

Is the requirement met, and what is actually needed?

A Two changes: permanent table, and Enterprise.

Transient tops out at 1 day whatever the edition, and Standard tops out at 1 day whatever the table type. Both limits bind.

B Yes, 30 days works anywhere.

What the engineer believes. The statement may not even error, but the retention will not take the requested value.

C Permanent is enough on Standard.

Half right. Standard permanent tables are still capped at 1 day; 90 days is the Enterprise figure.

D Business Critical required.

Business Critical adds encryption control, private connectivity and failover. It changes nothing about Time Travel retention.

WHERE THIS CAME FROM

Modules 2 and 4 together. Two independent ceilings, and fixing only one of them leaves you at 1 day.

Where each question came from

QUESTION	MODULE TO REVISIT IF YOU MISSED IT
1. The 60-second minimum	Module 1 - three layers, and the billing meters
2. ACCESS_HISTORY and masking	Module 2 - editions, the Enterprise cluster
3. Dropped column, unchanged storage	Module 3 - micro-partitions and immutability
4. Sharing a view	Module 4 - secure is a property, not a type
5. Enforced constraints	Module 4 - the other four table types
6. SnowCD	Module 5 - drivers, connectors and the SQL API
7. Cortex granted to PUBLIC	Module 6 - who is allowed to use Cortex
8. Thirty days of Time Travel	Modules 2 and 4 together

ON THE EXAM

Two or more misses clustered in one module is a signal. Two misses spread across six is normal and not worth chasing.

What this drill was really testing

- **Constraints that stack.** Question 8 had two independent ceilings. Fixing one is the commonest way to lose the mark.
- **True statements that answer nothing.** Questions 1 and 6 both offered one. Check relevance, not just accuracy.
- **Defaults nobody chose.** Cortex on PUBLIC, permanent as the default table type, load order as natural clustering.
- **The phrase that decides it.** 'Enforced by the database', 'shown the relevant passages', 'without changing who owns the writes'.

> ON THE EXAM
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> Next: Domain 2. Account management and governance, at 20%. Starting with the object hierarchy and RBAC.